```
import pandas as pd
import numpy as np
import seaborn as sns
```

```
df1 = sns.load_dataset("tips")
df1.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with df1 New interactive sheet

```
# create pivota table

pivot_table = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill")
pivot_table
```

```
/tmp/ipykernel_311/3586843793.py:3: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill")
```

smoker	Yes	No
sex		
Male	22.284500	19.791237
Female	17.977879	18.105185

Next steps: Generate code with pivot_table New interactive sheet

```
# aggfun

pivot_table1 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc="sum")
pivot_table1
```

```
/tmp/ipykernel_311/1139982709.py:3: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table1 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc="sum")
```

smoker	Yes	No
sex		
Male	1337.07	1919.75
Female	593.27	977.68

Next steps: Generate code with pivot_table1 New interactive sheet

```
pivot_table2 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc="count")
pivot_table2
```

```
/tmp/ipykernel_311/3627684940.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table2 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc="count")
```

smoker	Yes	No
sex		
Male	60	97
Female	33	54

Next steps: Generate code with pivot_table2 New interactive sheet

```
pivot_table3 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean",
pivot_table3
```

```
/tmp/ipykernel_311/791935718.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table3 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
/tmp/ipykernel_311/791935718.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table3 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
/tmp/ipykernel_311/791935718.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table3 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
```

	mean		sum		count	
smoker	Yes	No	Yes	No	Yes	No
sex						
Male	22.284500	19.791237	1337.07	1919.75	60	97
Female	17.977879	18.105185	593.27	977.68	33	54

Next steps: Generate code with pivot_table3 New interactive sheet

```
pivot_table4 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean",
pivot_table4
```

```
/tmp/ipykernel_311/766013015.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table4 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
/tmp/ipykernel_311/766013015.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table4 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
/tmp/ipykernel_311/766013015.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table4 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
/tmp/ipykernel_311/766013015.py:1: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table4 = df1.pivot_table(index = "sex",columns = "smoker",values = "total_bill",aggfunc=["mean","sum","count"
```

	mean		sum		count		std	
smoker	Yes	No	Yes	No	Yes	No	Yes	No
sex								
Male	22.284500	19.791237	1337.07	1919.75	60	97	9.911845	8.726566
Female	17.977879	18.105185	593.27	977.68	33	54	9.189751	7.286455

Next steps: Generate code with pivot_table4 New interactive sheet

```
pivot_table5 = df1.pivot_table(index = "sex",columns = "smoker",values = ["total_bill","tip","size"])
pivot_table5
```

```
/tmp/ipykernel_311/3244585775.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table5 = df1.pivot_table(index = "sex",columns = "smoker",values = ["total_bill","tip","size"])
```

	size		tip		total_bill	
smoker	Yes	No	Yes	No	Yes	No
sex						
Male	2.500000	2.711340	3.051167	3.113402	22.284500	19.791237
Female	2.242424	2.592593	2.931515	2.773519	17.977879	18.105185

Next steps: Generate code with pivot_table5 New interactive sheet

```
pivot_table6 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = "total_bill")
pivot_table6
```

```
/tmp/ipykernel_311/3884435363.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table6 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = "total_bill")
```

	day	Thur		Fri		Sat	Sun
	time	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner
sex	smoker						
Male	**Yes**	19.171000	NaN	11.386667	25.892	21.837778	26.141333
	No	18.486500	NaN	NaN	17.475	19.929063	20.403256
Female	**Yes**	19.218571	NaN	13.260000	12.200	20.266667	16.540000
	No	15.899167	18.78	15.980000	22.750	19.003846	20.824286

Next steps: Generate code with pivot_table6 New interactive sheet

```
pivot_table7 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bil]
pivot_table7
```

```
/tmp/ipykernel_311/1827133228.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table7 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bill","tip","size
```

		size						tip					
	day	Thur		Fri		Sat	Sun	Thur		Fri		Sat	Sun
	time	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner
sex	smoker												
Male	**Yes**	2.300000	NaN	1.666667	2.4	2.629630	2.600000	3.058000	NaN	1.90	3.246	2.879259	3.5213
	No	2.500000	NaN	NaN	2.0	2.656250	2.883721	2.941500	NaN	NaN	2.500	3.256563	3.1153
Female	**Yes**	2.428571	NaN	2.000000	2.0	2.200000	2.500000	2.990000	NaN	2.66	2.700	2.868667	3.5000
	No	2.500000	2.0	3.000000	2.0	2.307692	3.071429	2.437083	3.0	3.00	3.250	2.724615	3.3292

Next steps: Generate code with pivot_table7 New interactive sheet

```
pivot_table8 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bill"
pivot_table8
```

```
/tmp/ipykernel_311/1845221415.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table8 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bill","tip","size
```

		size						tip						to
	day	Thur		Fri		Sat	Sun	Thur		Fri		Sat	Sun	Th
	time	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner	Lu
sex	smoker													
Male	**Yes**	2.300000	NaN	1.666667	2.4	2.629630	2.600000	5.00	NaN	2.20	4.73	10.00	6.5	19
	No	2.500000	NaN	NaN	2.0	2.656250	2.883721	6.70	NaN	NaN	3.50	9.00	6.0	36
Female	**Yes**	2.428571	NaN	2.000000	2.0	2.200000	2.500000	5.00	NaN	3.48	4.30	6.50	4.0	13
	No	2.500000	2.0	3.000000	2.0	2.307692	3.071429	5.17	3.0	3.00	3.25	4.67	5.2	38

Next steps: Generate code with pivot_table8 New interactive sheet

```
pivot_table9 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bil]
pivot_table9
```

```
/tmp/ipykernel_311/1938951954.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table9 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bill","tip","size
```

		size						tip						to
	day	Thur		Fri		Sat	Sun	Thur		Fri		Sat	Sun	Th
	time	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner	Lu
sex	smoker													
Male	**Yes**	2.300000	NaN	1.666667	2.4	2.629630	2.600000	2.00	NaN	1.58	1.50	1.00	1.50	19
	No	2.500000	NaN	NaN	2.0	2.656250	2.883721	1.44	NaN	NaN	1.50	1.25	1.32	36
Female	**Yes**	2.428571	NaN	2.000000	2.0	2.200000	2.500000	2.00	NaN	2.00	1.00	1.00	3.00	13
	No	2.500000	2.0	3.000000	2.0	2.307692	3.071429	1.25	3.0	3.00	3.25	1.00	1.01	38

Next steps: Generate code with pivot_table9 New interactive sheet

```
# mergin

pivot_table6 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = "total_bill'
pivot_table6
```

```
/tmp/ipykernel_311/785692504.py:3: FutureWarning: The default value of observed=False is deprecated and will change t
  pivot_table6 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = "total_bill",margins=True
```

	day	Thur		Fri		Sat	Sun	All
	time	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner	
sex	smoker							
Male	**Yes**	19.171000	NaN	11.386667	25.892000	21.837778	26.141333	22.284500
	No	18.486500	NaN	NaN	17.475000	19.929063	20.403256	19.791237
Female	**Yes**	19.218571	NaN	13.260000	12.200000	20.266667	16.540000	17.977879
	No	15.899167	18.78	15.980000	22.750000	19.003846	20.824286	18.105185
All		17.664754	18.78	12.845714	19.663333	20.441379	21.410000	19.785943

Next steps: Generate code with pivot_table6 New interactive sheet

```
pivot_table9 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bil]
pivot_table9
```

```
/tmp/ipykernel_311/3973333683.py:1: FutureWarning: The default value of observed=False is deprecated and will change
  pivot_table9 = df1.pivot_table(index = ["sex","smoker"],columns = ["day","time"],values = ["total_bill","tip","size
```

		size							tip			...		tota]
	day	Thur		Fri		Sat	Sun	All	Thur		Fri	...	All	Thur
	time	Lunch	Dinner	Lunch	Dinner	Dinner	Dinner		Lunch	Dinner	Lunch	...		Lunch
sex	smoker													
Male	**Yes**	2.300000	NaN	1.666667	2.400000	2.629630	2.600000	2.500000	2.00	NaN	1.58	...	1.00	191.
	No	2.500000	NaN	NaN	2.000000	2.656250	2.883721	2.711340	1.44	NaN	NaN	...	1.25	369.
Female	**Yes**	2.428571	NaN	2.000000	2.000000	2.200000	2.500000	2.242424	2.00	NaN	2.00	...	1.00	134.
	No	2.500000	2.0	3.000000	2.000000	2.307692	3.071429	2.592593	1.25	3.0	3.00	...	1.00	381.
All		2.459016	2.0	2.000000	2.166667	2.517241	2.842105	2.569672	1.25	3.0	1.58	...	1.00	1077.

5 rows × 23 columns